HARD · GEOMETRY AND TRIGONOMETRY

Geometry and Trigonometry

10

10 questions · ~15 min

SAT QUESTION BANK

2026-05-29

Q1 GEOMETRY AND TRIGONOMETRY

A circle has center G, and points M and N lie on the circle. Line segments MH and NH are tangent to the circle at points M and N, respectively. If the radius of the circle is 168 millimeters and the perimeter of quadrilateral $GMHN$ is $3{,}856$ millimeters, what is the distance, in millimeters, between points G and H?

A. 168

B. 1,752

C. 1,760

D. 1,768

Q2 GEOMETRY AND TRIGONOMETRY

$$RS = 440$$

$$ST = 384$$

$$TR = 584$$

The side lengths of right triangle RST are given. Triangle RST is similar to triangle UVW, where S corresponds to V and T corresponds to W. What is the value of $\tan W$?

A. $\frac{48}{73}$

B. $\frac{55}{73}$

C. $\frac{48}{55}$

D. $\frac{55}{48}$

Q3

GEOMETRY AND TRIGONOMETRY

In triangle ABC, the measure of angle B is $90°$ and BD is an altitude of the triangle. The length of AB is 15 and the length of AC is 23 greater than the length of AB. What is the value of $\frac{BC}{BD}$?

A. $\frac{15}{38}$

B. $\frac{15}{23}$

C. $\frac{23}{15}$

D. $\frac{38}{15}$

Q4

GEOMETRY AND TRIGONOMETRY

A hemisphere is half of a sphere. If a hemisphere has a radius of 27 inches, which of the following is closest to the volume, in cubic inches, of this hemisphere?

A. 1,500

B. 6,100

C. 30,900

D. 41,200

Q5 GEOMETRY AND TRIGONOMETRY

A circle in the xy-plane has its center at $(19, 18)$ and has a radius of $9k$. Which equation represents this circle?

A. $(x - 19)^2 + (y - 18)^2 = 81k$

B. $(x - 19)^2 + (y - 18)^2 = 81k^2$

C. $(x - 19)^2 + (y - 18)^2 = 9k$

D. $(x - 19)^2 + (y - 18)^2 = 9k^2$

Q6

GRID-IN GEOMETRY AND TRIGONOMETRY

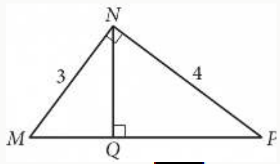
A right triangle has legs with lengths of 24 centimeters and 21 centimeters. If the length of this triangle's hypotenuse, in centimeters, can be written in the form $3\sqrt{d}$, where d is an integer, what is the value of d?

STUDENT-PRODUCED RESPONSE

.	/ .	/ .	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q7 GEOMETRY AND TRIGONOMETRY



In the figure above, what is the length of $\overline{NQ}$?

A. 2.2

B. 2.3

C. 2.4

D. 2.5

Q8

GRID-IN GEOMETRY AND TRIGONOMETRY

A right circular cone has a height of 22 centimeters cm and a base with a diameter of 6 cm. The volume of this cone is $n\pi$ cm^3. What is the value of n?

STUDENT-PRODUCED RESPONSE

.	/ .	/ .	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

$$x^2 + 20x + y^2 + 16y = -20$$

The equation above defines a circle in the xy-plane. What are the coordinates of the center of the circle?

A. $(-20, -16)$

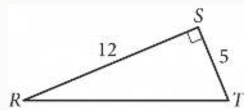
B. $(-10, -8)$

C. $(10, 8)$

D. $(20, 16)$

Q10

GRID-IN GEOMETRY AND TRIGONOMETRY



In triangle RST above, point W (not shown) lies on $\overline{RT}$. What is the value of $\cos(\angle RSW) - \sin(\angle WST)$?

STUDENT-PRODUCED RESPONSE

.	/ .	/ .	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.